

A Remuneration Model of Energy Community Members in Italy

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Abstract—The Directive (EU) 2018/2001 (RED II) requires Member States to authorize the constitution of consumers as self-consumers of renewable electricity by ensuring them non-discriminatory and disproportionate treatment. Member States must establish a favorable framework for promoting and facilitating the development of self-consumption, including in collective form (art. 21). According to the Directive, the granting of rights to renewable energy self-consumers who act collectively allows to increase the energy efficiency of households and helps to combat energy poverty by reducing consumption and supply tariffs. In Italy, the decree n. 162/2019 and then amended with Decree 199/2021 has partially enacted the RED II Directive. On the other hand, it must be observed that after more than 2 years the number of energy communities in Italy is still limited. One of the main barriers is related to the fact that there is no clear indication about how to share the incentive among the REC's members. In this paper, an effective model for the remuneration of the energy community members is proposed and an experiment is carried out in a laboratory environment, supported by Prycam Home devices.

Index Terms—Energy Community, Remuneration, Business models.

I. INTRODUCTION

According to the European regulatory framework, a Renewable Energy Community is defined as a "legal entity that is based on open and voluntary participation, it is autonomous and controlled by shareholders or members located in the proximity of renewable energy plants belonging to the community itself. The members may be physical persons, companies or local authorities, the main objective of the REC is not to produce financial profits but to provide environmental, economic and social benefits to its members or the local areas in which it operates" [1]. This definition recalls the concept of "proximity", which in the Italian legislation has

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been implemented by saying that producers, prosumers and consumers must stay downstream the same MV/LV substation. In many EU countries, however, the EC implementation by law is not yet a reality [2]. Even if many projects such as Energent, Ecopower, Wasewind, Enbro are running, collective energy actions aligned with the Clean Energy Package are still in early stage in Belgium. Both Renewable Energy Communities and Citizens energy communities (RECs and CECs) as defined in the EU directives are not yet transposed into national legislation in Croatia. There is a lack of a public consultation on the new electricity market law transposing the REC and, with it, the CEC into national legislation. As a result, collective self-consumption is not yet possible based on current Croatian regulatory framework. Latvia also does not have a transposition law for energy communities, but the National Action Plan for Energy by 2030 mentions 50% of installed generation capacity from renewable sources by 2030 as a target. To achieve it, the direction is to promote economically feasible self-consumption and self-generation involving society at large in energy production. Estonia is slightly more advanced with respect to EC regulatory transposition, as apartment associations can take over this task and there is a regulatory framework supporting such associations for all shared services in a building. In this case, with specific reference to electricity collective consumption, the main business model focuses on two issues:

- Collective consumption of electricity is cheaper for single consumers because the more electricity is consumed the cheaper it is. As a result, the savings come from cheaper electricity tariff for larger amounts of purchased electrical energy.
- There is a potential for collective self-consumption of locally produced PV based electricity. The PV produced electricity can only be sold at the market price, making self-consumption more interesting for end consumers.

However the business model for electricity has not been attractive till now, as electricity is still cheap in Estonia, although the Ukraine crisis and the consequent diversification of gas suppliers may impact prices of electricity. Before the issuing of REDII, since January 2018, Greece has regulated new types of energy cooperation within a wider framework of "Social and Solidarity Economy". According to the Greek law, an energy community can operate in a number of different activities else than those foreseen by REC and CEC. Two types of cooperatives can be instantiated: non-profit and for-profit. The cooperatives can operate on the heating and electricity market based on renewables or high-quality cogeneration, and are open to any entity other than large companies. Privileged treatment in production licensing and connection procedures are ensured to energy communities in Greece. The islands of Sifnos and Kithos are good examples of a wide and full application of energy community models [3]. In Germany [4], different models are available since many years for energy cooperation. There are essentially 3 different models: energy cooperatives, collective self-consumption, citizen energy companies. The first model entails the creation of cooperatives, that can share profits and losses so that surpluses are reinvested to support its members and the community. Financial profits are distributed amongst the cooperative members through dividends and/or reduced energy prices. In the case of collective self-consumption, energy sharing among consumers is still not possible. The PV plant owner can sell electricity to the apartment owners in the same building and will receive an incentive based on the self-consumed electricity [5]. In Italy, two recent decrees have transposed the EU norms for Renewable EC (decree 8/2020 and decree 199/2021), introducing the following two models for collective energy actions:

- collective self-consumers (CSC), and
- renewable energy communities (REC).

The first model (CSC) is applicable to condominiums, in this case, members can thus be natural persons or commercial entities, not having as core business generation and energy exchange, and that are located in the same building or condominium. In this way, low-income and vulnerable residents can be included. The second model (REC) is applicable to associations of natural persons, small and medium enterprises, local/regional authorities (e.g. municipal administrations), and private companies. The main aim of a REC is to grant environmental, economic, or social benefits to its members/ local area, rather than financial profits. In both cases, generation plants (individually not exceeding 0,2 MW) and other members of the REC must stay downstream the same secondary substation. Such community can also carry out aggregation activities and act as a balancing service provider. In all illustrated cases where REC are possible by law, there is no explicit incentive for maximizing self-consumption from the REC components. Or at least this is not regulated by law and thus publicly available. The issue of devising a way to distribute the benefit among the REC members is dealt with in [6], where business models to guarantee a fair distribution and

to prevent the penalization of someone are proposed. The authors have developed a program that simulates different business models to be proposed to prosumers in multifamily residential buildings, through which the economic return of each participant in the energy community is evaluated by considering the cost of energy purchased from the grid, from the sale of excess production, and the economic revenues from self-consumption. The latter are divided proportionally to each user's self-consumption. This last step is one of the main barriers to the deployment of this new reality. In fact, there is no clear indication on how to divide the incentive among the REC members. To date, as indicated in [7], the incentive is given to the REC manager, but no guidelines are given on how to distribute the incentive among the members. Therefore, in this paper, an effective model for the remuneration of energy community members is proposed and experimentation is carried out in a laboratory environment, with the support of Prycam Home devices.

II. REMUNERATION SYSTEMS FOR PARTICIPATIVE ENERGY COMMUNITIES IN ITALY

In Italy, RECs and CSCs are not having much success. Obstacles include bureaucracy, the cost of investing in renewable generation, and the lack of a clear and transparent remuneration system for REC members.

The economic benefits received by RECs are derived from shared electricity and energy fed into the grid that is not shared. In particular, the GSE grants a premium tariff (PT) of €100 for each MWh of shared energy. The total benefit for the REC for a single day:

$$B_{REC,d} = PT * \sum_{h=1}^{24} E_{S,h} \quad (1)$$

Where:

- $E_{S,h}$ is the hourly shared energy expressed in MWh;

Shared energy is defined as the minimum, calculated on an hourly basis, between the electricity injected into the grid by renewable-fueled generation facilities and the electricity consumed through Points of Connection (PoC) that are part of CSC or REC:

$$E_{S,h} = \min \left[\sum_{y=1}^n E_{gen,y}; \sum_{i=1}^n E_{cons,i} \right] \quad (2)$$

Where:

- $E_{gen,y}$ is the energy actually generated and injected into the grid through the y -th active PoC expressed in MWh;
- $E_{cons,i}$ is the energy actually consumed through the i -th passive PoC expressed in MWh;

The aforementioned contributions assessed on the basis of shared energy, expressed in €, are granted by the GSE (Gestore Servizi Energetici) after receiving electricity measurements from grid operators and the data necessary for their validation.

In the event that the GSE has available all the necessary and valid measurements for calculation transmitted by the grid operators, it performs the complete calculation on a monthly basis of the following economic items:

- with reference to shared electricity, for each configuration, the contribution due expressed in €.
- with reference to the electricity fed into the grid, for each configuration, the value of the energy withdrawn expressed in €.

If, instead, the measurement data were only partially available, the GSE will proceed with a partial calculation using only the measurement data transmitted by the grid operators and considered valid by the GSE for the reference period, and then perform a full recalculation only when it has all valid data available for the reference period. The incentive is granted by the GSE to the REC Referent on a monthly basis.

While the calculation of the incentive is well defined, there is still no explicit way to distribute the incentive among community members. A number of models are currently available from the literature for recognizing a benefit to individual EC components. However, not in all legislations the self-consumption such as it is defined in Italy is considered. In all cases, the easiest model considers EC components as members of a cooperative/company, thus simply benefit uniformly from being part of an associative form to whom a shared financial benefit is granted. In this way, each member would get an equal percentage of what is gained every year. However, without a feedback or a coordination, there is a risk to overreact leading for example to every one charging their electric vehicle at the same time and possibly increase the electric load above the local PV production or on the opposite to have very small economic benefit because people are not aware about the mechanism and their consumptions. In [8], two REC CSC models are presented, one in which the distribution grid connecting the members of the community is privately owned and only one PoC is present (physical model) and another model where each apartment or business or entity has a single PoC and thus a single meter (virtual model). As indicated also in [7], the virtual model appears to be the most common, and for this reason, in this work, some algorithms are proposed for distributing the monthly benefit paid by the GSE in order to incentivize each member to do their best to increase energy sharing.

A. Remuneration model 1

The model considers that both producers and consumers contribute to gaining the benefit. For this reason, the overall benefit is shared in two parts, one for generation units and one for consumers units. In each hour of the year, for the purely passive i -th consumer, the benefit for each hour h of the generic day d can be written as follows:

$$B_{d,i,h} = \alpha * B_{REC,d,h} * \frac{E_{d,i,h}}{\sum_{i=1}^n E_{d,i,h}} \quad (3)$$

For the y -th producer, the benefit for each hour h of the generic day d can be written as follows:

$$B_{d,y,h} = \beta * B_{REC,d,h} * \frac{E_{d,y,h}}{\sum_{y=1}^m E_{d,y,h}} \quad (4)$$

$$\alpha = \frac{\sum_{i=1}^n P_{rtd,i}}{\sum_{i=1}^n P_{rtd,i} + \sum_{y=1}^m P_{rtd,y}} \quad (5)$$

$$\beta = \frac{\sum_{y=1}^m P_{rtd,y}}{\sum_{i=1}^n P_{rtd,i} + \sum_{y=1}^m P_{rtd,y}} \quad (6)$$

Where:

- $B_{REC,d,h}$ is the total benefit evaluated as for (1) at hour h for the generic day d ;
- $E_{d,i,h}$ is the energy consumed at hour h by the i -th consumer during the generic day d ;
- $E_{d,y,h}$ is the energy produced at hour h by the y -th producer during the generic day d ;
- $P_{rtd,i}$ is the rated power or the capacity available for the i -th consumer;
- $P_{rtd,y}$ is the rated generation power for the y -th producer;

In case a prosumer is also part of the REC, the benefit for each hour h of the generic day d will be considered as the sum of (3) and (4).

Through the coefficients α and β , the total benefit gained from the shared energy is divided equally between the two groups of consumers and producers if production and consumption capacities are equal. Otherwise, it is divided proportionally to production and consumption capacities. This not only encourages passive users to consume during production hours, but also optimizes the REC's environmental benefit, which is the better the more production and consumption are balanced.

B. Remuneration model 2

This model considers for each passive user a minimum consumption threshold in each hour of the day. In this way each member also has in principle a cap on the maximum benefit he/she can take out of the total benefit granted by the GSE evaluated as in (1). Specifically, if in the hour h -th hour of the generic day d the energy shared was equal to $E_{S,d,h}$ and the REC consists of n passive users, the minimum threshold for the i -th passive user to receive a share of the benefit is:

$$S_{d,i,h} = \frac{P_{rtd,i}}{\sum_{i=1}^n P_{rtd,i}} * E_{S,d,h} \quad (7)$$

Where:

- $P_{rtd,i}$ is the rated power of the i -th passive user belonging the REC;
- $E_{S,d,h}$ is the energy shared during the h -th hour of the generic day d evaluated as in (2).

If a user during the h -th hour reaches or exceeds the minimum threshold he/she will be granted the incentive. On the other hand, if a user during the h -th hour does not reach the minimum threshold then he/she will not be remunerated and the incentive will be distributed to the other members. The total benefit for each consumer and producer is evaluated in the

same way as for Model 1, but in this case, while producers will always be granted a contribution proportional to the energy produced, consumers will be granted a contribution only in the hours when they exceeded the threshold.

As with the previous model, in the case of prosumer, the benefit will be the sum of the two contributions above.

While in the previous model users always receive the benefit because it is assessed on the basis of the energy consumed in the h -th hour, in this model the benefit is obtained only by users who exceed a certain threshold defined on the basis of their rated power or available capacity. This provides a greater encouragement to consume during production hours.

III. APPLICATION

With the aim of testing the remuneration models described in the previous section, a REC was emulated within the Smart and Microgrid Lab (SMGLab) at the University of Palermo. A testbed for emulating household users that can be classified as smart prosumers is installed within the laboratory. The testbed was built with the aim of emulating a smart user, i.e., optimally managing production and consumption for providing ancillary services to the grid or participating in self-consumption of an energy community. It allows emulating at the same time three different passive users and one active user connected downstream of the same node of a Low Voltage (LV) grid. The maximum power of each user, both active and passive, is 3 kW. Fig. 1 shows the complete test bench installed inside the laboratory, from which all system components can be seen.



Fig. 1. Photo of the test bench.

The main components are:

- 1) power generation system: for simulation of generation from a photovoltaic system;
- 2) switchable resistors: for simulation of step-variable passive loads;
- 3) controllable electric motor: for simulation of active loads;
- 4) storage system: for simulating energy exchange with the grid;
- 5) an energy management system: for coordinated control of all components;

- 6) intelligent monitoring and metering system: for monitoring, measurement of load and generation profiles, and communications with the energy management system and the control PC;
- 7) protection and isolation transformer: for protection against electric shock and overcurrent.

The power generation system aims to emulate a renewable energy plant, in this case photovoltaic, exchanging energy with the storage system and loads, represented by the switchable resistors (passive loads) and the electric motor (active load). The correct management of the inverter, storage, and loads enables the typical load or generation profiles of a prosumer being part of a microgrid or emulating a small energy community consisting of 3 passive users and a photovoltaic power generation system. The consumption or production of the system and all its components, besides being monitored through energy meters is also monitored through PRY-CAM HOME devices [9]. PRY-CAM HOME is a monitoring system developed by Prysmian Electronics S.r.l. for LV applications based on PRY-CAM technology and consists of a device used to measure the main parameters of a domestic low-voltage electrical system. It can be defined as a Pro-Sumer device because it can be used by both experts and domestic end-users with different functions depending on the user's skills, but with the same purpose: to verify the correct functionality of the electrical system and, therefore, its safety. The device is usually placed in the home electrical panel below the energy meter; it measures the electrical parameters of the system, stores them on a cloud platform and makes them available for monitoring via a mobile App. The App makes it possible to monitor the main parameters of the electrical system (both historical and real-time) and alert the user in case of unusual behavior of the system (unsafe device connected, failure in the ground connection, no power supply or high temperature in the electrical panel, need for system maintenance, etc.). Tab. 1 shows the main quantities measured and sent by PRY-CAM HOME to the cloud platform.

TABLE I
PRY-CAM HOME MEASURED QUANTITIES.

Value to send		Types of value to send (1/min)
Phase-to-neutral Voltage	V_{FN}	$V_{FNmin}; V_{FNavg}; V_{FNmax}$
Phase current	I_F	$I_{Fmin}; I_{Favg}; I_{Fmax}$
Differential current	$I_{\Delta n}$	$I_{\Delta navg}$
Active power	P	P_{avg}
Reactive power	Q	Q_{avg}
Current to Voltage Phase shift	$\cos\phi$	$\cos\phi_{avg}$
Frequency	F	$F_{min}; F_{avg}; F_{max}$
Temperature	T	T_{avg}
Energy	E	E

For the emulation of the REC, four PRY-CAM HOME devices were installed inside the switchboard of the test bench. Specifically, one was installed to monitor the energy produced by the photovoltaic simulator, three were installed to monitor the emulated energy consumption through the appropriate

control of the resistors. The consumption profiles of the three passive users are emulated by controlling the closing/opening of contactors installed inside the switchboard, which are controlled through Matlab by implementing a MonteCarlo method for domestic load profile generations. Fig. 2 shows the load profiles for a summer day emulated for the three users through the test bench and measured by the PRY-CAM HOME. While, Fig. 3 shows the generation profile of two summer days emulated through the PV simulator.

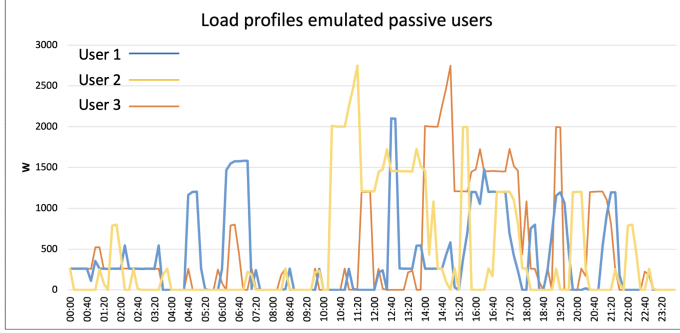


Fig. 2. Load profiles emulated through the test bench.

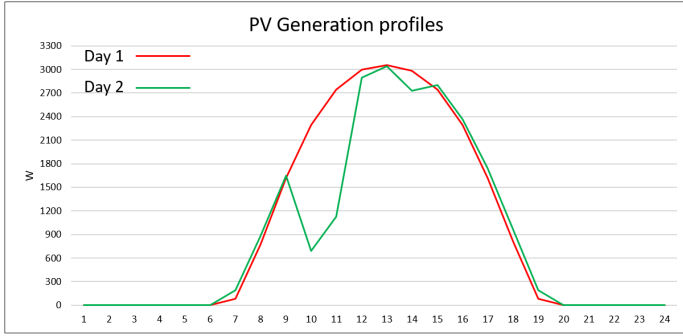


Fig. 3. Generation profiles emulated through the test bench.

IV. RESULTS

The load and generation profiles emulated through the test bench proper control are measured by the PRY-CAMs and sent to the cloud. Through the Matlab MQTT library, the measured data from PRY-CAMs can be obtained in order to evaluate the remuneration models exposed in Section II. Fig. 4 shows the REC architecture emulated.

Using the same load profiles shown in fig. 2, the two remuneration models were evaluated for each of the generation profiles shown in fig. 3. Applying the definition in (2) the total shared energy in the two cases is 24 kWh/day and 21 kWh/day. So the total benefit per day for the REC will be 2,4 € and 2,1 €, respectively.

The following Table II shows the results obtained by applying remuneration model 1 to the emulated REC.

While the following Table III shows the results obtained by applying remuneration model 2.

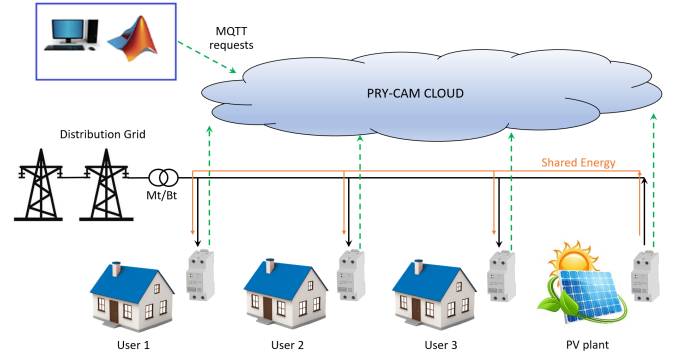


Fig. 4. REC architecture emulated using the test bench.

TABLE II
RESULTS BY APPLYING MODEL 1

Remuneration Model 1				
Day 1	α	β	E_s [kWh/day]	B [€/day]
	0,75	0,25	24,07	2,41
	B_{us1} [€/day]	B_{us2} [€/day]	B_{us3} [€/day]	B_{prod} [€/day]
Day 2	0,65	0,35	0,78	0,63
	α	β	E_s [kWh/day]	B [€/day]
	0,75	0,25	21,25	2,14
	B_{us1} [€/day]	B_{us2} [€/day]	B_{us3} [€/day]	B_{prod} [€/day]
	0,51	0,32	0,81	0,50

As can be seen by comparing the results, using model 2 passive user 1 gained less than using model 1. In fact, this result depends on the use of the minimum threshold for accessing the benefit. Model 2 compared to model 1, encourages users to consume during the PV system production hours. This result can be improved through the use of an energy management system that distributes household loads during production hours.

The profit for the producer remains the same as it depends on the installed capacity. For the emulated energy community, considering an investment cost for PV system installation of 4000 €, annual producibility of 4500 kWh, the annual shared energy corresponds to the producibility of the system and the selling price of the energy injected into the grid is 0.06 €, the payback time for both models is about 10 years. Whereas, if the generation capacity is equal to the consumption of the community ($\alpha = \beta = 0.5$), the payback time for producers is reduced to about 7 years.

The payback time does not change with the remuneration model, but using model 2, in addition to incentivizing passive users to consume during production hours, producers are further incentivized to install generation capacity that is as equal as possible to the REC's total consumption, thus maximizing both user's and environmental benefits.

V. CONCLUSION

This paper proposes a remuneration model for users who are part of a REC that allows the benefit earned to be divided equally to the contribution provided and encourages passive users to consume during production hours. Using PRY-CAM HOME devices and the testbed hosted within the SMGLab at

TABLE III
RESULTS BY APPLYING MODEL 2

Remuneration Model 2				
Day 1	α	β	Es [kWh/day]	B [€/day]
	0,75	0,25	24,07	2,41
	B_{us1} [€/day]	B_{us2} [€/day]	B_{us3} [€/day]	B_{prod} [€/day]
Day 2	0,25	0,43	0,90	0,63
	α	β	Es [kWh/day]	B [€/day]
	0,75	0,25	21,25	2,14
	B_{us1} [€/day]	B_{us2} [€/day]	B_{us3} [€/day]	B_{prod} [€/day]
	0,19	0,35	0,93	0,50

the University of Palermo, it was possible to test the proposed model through the emulation of a small REC consisting of three passive users and a photovoltaic system. The developed model shares the incentive earned by the community based on each user's contribution to the shared energy. In this way, each EC member is well encouraged to improve his/her behavior and thus increases the benefit. Future work includes the implementation of an energy management system with the aim of maximizing the shared energy and thus the benefit to the entire community.

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